

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Digital Logic

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

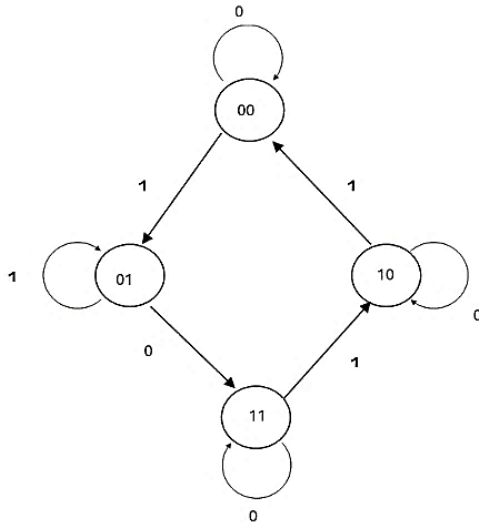
Attempt all the questions.

1. a) Differentiate analog system and digital system. Which system do you prefer analog or digital and why? 5
b) Perform the following conversions: 5
 - i. $(73)_8 = (?)_5$
 - ii. $(101101)_2 = (?)_{\text{Gray}}$
 - iii. $(101011010)_2 = (?)_8, (?)_{16}$
 - iv. Find x if $(201)_x = (143)_8$
- c) Perform subtraction using 1's complement method. 5
 - i. $(101101-1010)_2$
 - ii. $(11011-111101)_2$
2. a) Why NAND and NOR gates are called Universal Gate? Implement NAND gate as Universal Gate. 5
b) Simplify using Boolean Algebra 5
 - i. $A + B[AC + B\{AC + (B + \bar{C})D\}]$
 - ii. $AB + A(B + C) + B(B + C)$
- c) Simplify the function using K-map $F = \sum(0,1,4,8,10,11,12)$ and $D = \sum(2,3,6,9,15)$. Also realize the simplified circuit using NOR gates. 5
3. a) What is a Full Adder? Show that Full Adder is the combination of two Half Adder and OR Gate. 1+7
b) Design a combinational circuit that has four inputs and two outputs. One of the output is high if majority inputs are high and the second output is high only when all inputs are of same type. 7
4. a) Design a three bit magnitude comparator that perform $A > B$, $A < B$ and $A = B$. 8
b) Realize $F(wxyz) = \sum(1,2,4,11,12,14,15)$ using $8 \times 1 \text{ MUX}$. 7

OR

What is decoder? Design 3×8 decoder circuit.

5. a) Realize the following state diagram into a sequential circuit using JK flipflop: 8



- b) Explain the JK flip-flop with the necessary tables and diagrams. 7
6. a) Define counter. Design a 3-bit synchronous up counter using T-flip flops. 8

OR

Define register and shift register. Explain the working of serial in serial out shift register with its circuit and timing diagram.

- b) Design a 4-bit arithmetic circuit with selection variables S_1 and S_2 which performs following eight different arithmetic operations: 7

S_1	S_0	C_{in}	F
0	0	0	A
0	0	1	A+1
0	1	0	A+B
0	1	1	A +B+1
1	0	0	A+B'
1	0	1	A+B'+1
1	1	0	A-1
1	1	1	A

7. Write short notes on: (**Any two**) 2×5
- Don't care condition
 - Output hazard races
 - Status Register